

Course syllabus for

Academic year 2025/2026 >

MCC185 - From quantum optics to quantum technologies

Från kvantoptik till kvantteknologier

Course syllabus adopted 2023-02-08 by Head of Programme (or corresponding)

Owner: [MPNAT](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: Second-cycle

Main field of study: Engineering Physics

Department: 59 - MICROTECHNOLOGY AND NANOSCIENCE

Course round 1

Teaching language: English

Application code: 18113

Open for exchange students: Yes

Block schedule: [C](#)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 		
0122 Examination 7,5 c Grading: TH	7,5 c						Tue 28/10-2025 am J,	Contact examiner,	Contact examiner

In programs

[MPNAT Nanotechnology, Year 1 \(compulsory elective\)](#)

[MPNAT Nanotechnology, Year 2 \(elective\)](#)

[MPPHS Physics, Year 1 \(compulsory elective\)](#)

[MPPHS Physics, Year 2 \(elective\)](#)

Examiner:

[Simone Gasparinetti](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for Master's level (second cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

FUF040 Quantum Physics or equivalent is required. The following courses are recommended: TIF101 Applied Quantum Physics and TIF290 Quantum Mechanics.

Aim

The course introduces how one can describe, manipulate, and detect quantum mechanical systems such as single atoms and photons, and how advances in the control and measurement of these systems are driving the so-called "second quantum revolution" through the four pillars of quantum technologies: quantum computation, quantum simulation, quantum communication, and quantum sensing. This revolution is being pushed forward by large research initiatives worldwide, such as the Wallenberg Centre for Quantum Technology (WACQT) in Sweden, of which Chalmers is the main node, the EU Quantum Flagship in Europe, and many more. The course gives an overview on this very active field of research and connects via lectures, exercise sessions, and a laboratory session in a state-of-the-art facility to ongoing research on quantum mechanical superconducting circuits, microwave photons, and optomechanical systems.

Learning outcomes (*after completion of the course the student should be able to*)

After the course the student should be able to:

- Understand the difference between classical and non-classical radiation
- Explain the properties of the Jaynes-Cummings model
- Use the Bloch equations to describe the dissipative dynamics of a quantum mechanical two-level system
- Compute the output state of simple quantum circuits composed of elementary single-qubit operations, entangling gates and measurements
- Have a basic knowledge of the leading architectures to build a quantum computer and their comparative advantages and disadvantages
- Understand the difference between a quantum computer and a quantum simulator, and discuss use cases for both
- Understand how quantum technology may threaten today's encryption keys and how secure communication can be established by quantum links
- Explain the standard quantum limit and how to break it
- Explain and experimentally perform manipulations and tomographic measurements of quantum states of a microwave resonator, assisted by a superconducting qubit

Content

In the first part of the course, we will study the foundations of quantum optics, that is, how matter (atoms) interacts with an electromagnetic field at the quantum level (photons). We will study both the semi-classical and the full quantum light-matter interaction and get to know different quantum states of light and their quantum optical description. In experiments that implement these building blocks of quantum optics, one can use a diverse set of physical systems, for example, atoms, trapped ions or artificial atoms such as superconducting microelectronic circuits that possess quantum mechanical properties like atoms.

In the second part, we will apply the formalism of quantum optics to various physical platforms including superconducting circuits, trapped ions, cold atoms, nitrogen-vacancy centers in diamond and mechanical resonators. Here the goal is to understand how quantum effects can be exploited to build novel devices and to use novel measurement techniques, that are key for all four pillars of quantum technology. Quantum computers allow to perform certain computations or simulations by using quantum algorithms that are faster than the corresponding classical algorithms. The development of quantum simulators can be traced back to R. Feynman's intuition that carefully engineered quantum systems could be used to efficiently simulate materials and molecules, potentially leading to breakthroughs in material science and chemistry. Quantum communication systems allow performing quantum key distribution over absolute safe channels and can connect quantum computers over large distances. Finally, quantum sensors take advantage of quantum phenomena such as state superposition and squeezing to build sensors, imaging systems, and metrological standards with unprecedented accuracy.

Part I

1. Building blocks of quantum mechanics and quantum optics
2. Photons: classical and non-classical states of radiation
3. Wigner functions and Wigner tomography
4. Atom-field interactions: Rabi oscillations and the Jaynes-Cummings Hamiltonian
5. Quantum decoherence and the Bloch equations
6. Readout of quantum information

Part II

1. Basics of quantum information processing
2. Overview of architectures for quantum information processing
3. Circuit quantum electrodynamics and its applications
4. Quantum simulation
5. Quantum communication
6. Microwave-to-optical transduction
7. Quantum sensing

Organisation

Lectures, exercises, home work, and a state-of-the art experiment with report writing

Literature

Lecture notes, hand-outs.

The following literature is good but not strictly necessary to acquire:

"Introductory Quantum Optics" Christopher Gerry and Peter Knight, Cambridge University Press, ISBN-10: 052152735X

"Quantum Computation and Quantum Information" Michael A. Nielsen and Isaac L. Chuang Cambridge University Press (2000) ISBN 0 521 63503 9. Can be found as an e-book in the library.

Examination including compulsory elements

The course examination will consist of: 4 obligatory hand-ins, lab report and exam. To pass the course, you need to obtain at least 40% of the points on the exam and participate in the lab and submit a written lab report. The grade will then be based on: exam (50%), hand-ins (35%) and lab report (15%).

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers about disability study support.